

1. Answer all the questions (1X6=6)

- (a) Find the average of all the prime numbers between 80 and 90.
- (b) Find HCF of a^3b^2 , a^2b^3 .
- (c) If $v = u + ft$, express t in terms of u, v, f .
- (d) Find the number of sides of a regular polygon whose One of the exterior angle is 40° .
- (e) Measure of four angles of a pentagon are $73^\circ, 97^\circ, 146^\circ, 138^\circ$. Find measure of fifth angle.
- (f) A triangle having the lengths of sides 13 cm, 14 cm and 15 cm. Find its perimeter.

2. Answer all the questions. (2X7=14)

- (a) In a wall clock it rings 5 times in 3 seconds, then what time it will take to ring 6 times?
- (b) Find the interest on Rs 750 for 3 years at a rate of simple interest 5%.
- (c) If $\left(a + \frac{1}{a}\right)^2 = 3$, then find $a^3 + \frac{1}{a^3}$
- (d) If $I = \frac{nE}{R + nr}$, then express n in terms of I, E, R, r .
- (e) If $x + \frac{1}{x} = 2$, then find x .
- (f) In triangle ABC side BC extend up to D. If $\angle ACD = 125^\circ$ and $\angle B = \frac{2}{3}\angle A$, then find $\angle A$.
- (g) In a parallelogram ABCD, $m\angle A = 55^\circ$, find $m\angle B$.

3. Answer any two. (7X2=14)

- (a) A
- (b) A
- (c) a

4. Factorize the following: (4X1=1)

$$x^3 - 12x - 16 \quad \text{or} \quad a^3 - 9b^3 - 3ab(a - b)$$

5. Find HCF of the following:

(4X1=4)

$$3a^3 - 12a^2 + 21a - 18 \text{ and } 6a^3 - 30a^2 + 60a - 48$$

$$\text{Or } 2p^3 - 5p^2 + 5p - 3 \text{ and } 3p^3 - 5p^2 + 5p - 2$$

6. Simplify the following:

(4X1=4)

$$\frac{a(a-b)^2 + 4a^2b}{ab+b^2} \div \frac{a^2-b^2}{ab} \times \frac{b(a+b)^2 - 4ab^2}{a^2-ab}$$

$$\text{Or } \frac{b-c}{a^2-(b-c)^2} + \frac{c-a}{b^2-(c-a)^2} + \frac{a-b}{c^2-(a-b)^2}$$

7. Solve the following:

(4X1=4)

$$\frac{x-a}{b+c} + \frac{x-b}{c+a} + \frac{x-c}{a+b} = 3$$

$$\text{Or } \frac{6x-7}{9x+6} - \frac{5(x-1)}{12x+8} = \frac{1}{12}$$

8. Solve the following:

(4X1=4)

$$2x - 3y = 8$$

$$\frac{x+y}{x-y} = \frac{7}{3} \quad \text{by Comparison method.}$$

$$\text{Or } \begin{array}{l} 7x - 3y = 31 \\ 9x - 5y = 41 \end{array} \quad \text{by method.}$$

9. A

10. A

11. A

12. A

13. A

14. A

15. A

16. a